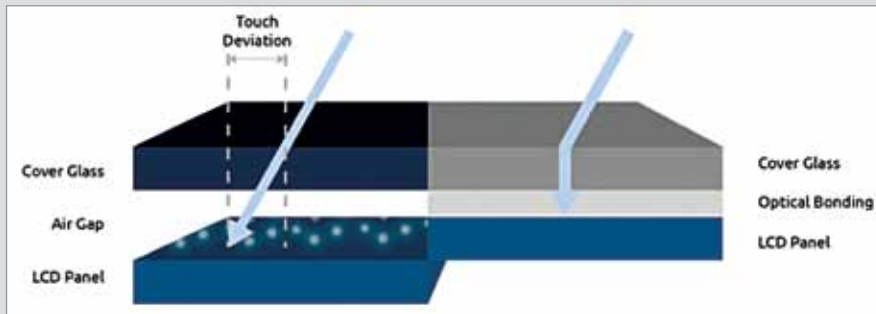




Optical bonding with cover glass (Credit: Newline Interactive)

Interface. Low-voltage differential signalling (LVDS) could be a popular choice for giant LCDs and peripherals in need of high bandwidth, like high-definition graphics and fast frame rates. It is most ordinarily utilised in automotive TFT applications. It's an excellent solution due to its high speed of data transmission while using low voltage.

Two wires carry the signal, with one wire carrying the precise inverse of its companion. The electrical field generated by one wire is neatly concealed by the opposite, creating much less interference to nearby wireless systems. At the receiver end, a circuit reads the difference (hence call Differential) in voltage between the wires. Therefore, this does not generate any noise or gets its signals scrambled by external noise. The interface consists of four, six, or eight pairs of wires, plus a pair carrying the clock and a few ground wires.

Wide viewing angle. Centre-stack displays should be visible to both driver and the passengers, including those on the rear seats. LCD display modules are designed to provide the best contrast and readability in one of the four directions, known as the Viewing Angle or Optimal Viewing Direction. These

four directions are laid out in the shape of a clock. The top view is at 12:00pm and the bottom view is at 6:00pm.

Touch and glass bonding.

Optical bonding is the method involving overlaying touchscreens, glass, or plastic cover lenses, adding EMI filters and other display upgrades to an LCD. A layer of glue is incorporated between the cover layer and display to fill

the air gap abandoned in a regular edge or gasket bonding. Optical bonding helps in enhancing clarity, viewability, reducing reflections, and making TFT more rugged, which are critical factors in automotive applications.

Wide temperature range. Temperature range can typically span from -40°C to 85°C. Such a high-temperature range is required because in summer ambient temperature adds up with the operational temperature of TFT, where normally in desert areas ambient temperature rises to 55°C in summer.

Colour depth. Higher-resolution displays might have to upgrade from 18-bit red, green, blue (RGB) to 24-bit RGB to attain a wider colour gamut.

Refresh rate and response time. Keeping away from slacks is basic for warning indicators and navigation functions like maps and traffic updates.

Power consumption. Low power consumption enables better fuel consumption and allows components to be placed in hot spots.

Lifetime and production support. Displays must support design and production cycles of a minimum of five years, extendable up to ten years because of vehicle warranties.

absorber. Shakes and shocks are less likely to damage the display and glass, making this an important benefit for transportation applications. In the unlikely event that the glass is damaged, shards of broken glass will remain stuck to the optical adhesive.

Reflection or glaring sunlight sometimes makes it difficult to read the displays. Any innovation introduced recently, or underway, that may solve this issue?

Reflective LCD and Progressive

Super View are the two technologies which are effective under high ambient light. In progressive super view technology, internal and external reflection is cut down, which results in a clear view without glare. And the beauty of this technology is that it happens without pumping more power from the backlight. This helps in more lifetime of the backlight and less power consumption.

Reflective LCD is another technology that uses ambient light to reflect in order to read, hence there

is more clarity under sunlight and very less power is needed. It is more beneficial for EV applications.

How do LCDs combat the continuous temperature changes throughout the day, especially in automobiles?

Automotive-grade LCDs have strict requirements. The LCD must remain working during the extreme environment, for example, Indian summertime. For example, our LCDs are tested for storage temperature of